

A data-driven approach to selection of critical process steps in the semiconductor manufacturing process considering missing and imbalanced data

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ABSTRACT

Semiconductor wafers are fabricated through sequential process steps. Some process steps have a strong relationship with wafer yield, and these are called critical process steps. Because wafer yield is a key performance index in wafer fabrication, the critical process steps should be carefully selected and managed. This paper proposes a systematic and data-driven approach for identifying the critical process steps. The proposed method considers troublesome properties of the data from the process steps such as imbalanced data, missing values, and random sampling. As a case study, we analyzed hypothetical operational data and confirmed that the proposed method works well.

1. Introduction

Semiconductor wafer fabrication processes consist of hundreds of complex processes, many of which are performed repeatedly, such as oxidation, photolithography, cleaning, etching, and planarization. These are “unit processes” in semiconductor wafer fabrication, and a “process step” is a unit of work of a process performed by a machine for each wafer. Usually, more than 500 process steps are required to fabricate semiconductor wafers. Fig. 1 shows the concept of process steps in semiconductor wafer fabrication.

Wafer yield is calculated as the ratio of good to total semiconductor chips within the wafer. Maintaining high yield through reliable and accurate quality control for wafer fabrication is a key success factor in the semiconductor industry [1]. The first task for yield improvement is selecting process steps that have a strong relationship with wafer yield, and these steps are called “critical process steps.” Once the critical process steps are identified, yield improvement can be performed by optimizing the critical process steps.

The conventional approach for selecting the critical process steps is evaluating the technical importance of each process step. The technical importance is usually based on a process engineer's knowledge and experience; thus, the process engineer's subjective judgement is involved in the evaluation process. The conventional approach requires significant human resources because the number of process steps is often large (e.g., more than 500 process steps), and evaluation is

conducted manually by process engineers.

Analyzing a large volume of operational data from semiconductor wafer fabrication lines is an attractive alternative to overcome the difficulties of the conventional approach. It is common to use operational data from manufacturing lines due to the development of information technologies such as IoT (Internet of things) and explosive growth of data generated from manufacturing lines [2]. Thus, it is increasingly important to analyze these data for process optimization.

Kadlec et al. [3] pointed out common properties of operational data such as missing values and data outliers. We have found three troublesome properties that are often encountered in semiconductor wafer fabrication line when analyzing process operation data. The first property is a high rate of missing values due to a low measurement rate. Once a wafer goes through the process steps, it is inspected by measurement equipment. Because of the time-consuming method and the limited capacity of the measurement equipment, only a small portion of wafers can be measured [4].

The second property is random sampling for measurement. For each process step, wafers are randomly selected for measurement (or inspection). Under these circumstances, wafers have an extremely low probability of complete measurement data for all of the process steps, which makes it difficult to investigate the correlation among them. For a simple case, suppose that we have five process steps (in practice, more than 500 process steps exist). When the measurement rate is 20% and wafers are randomly and independently selected, the probability of

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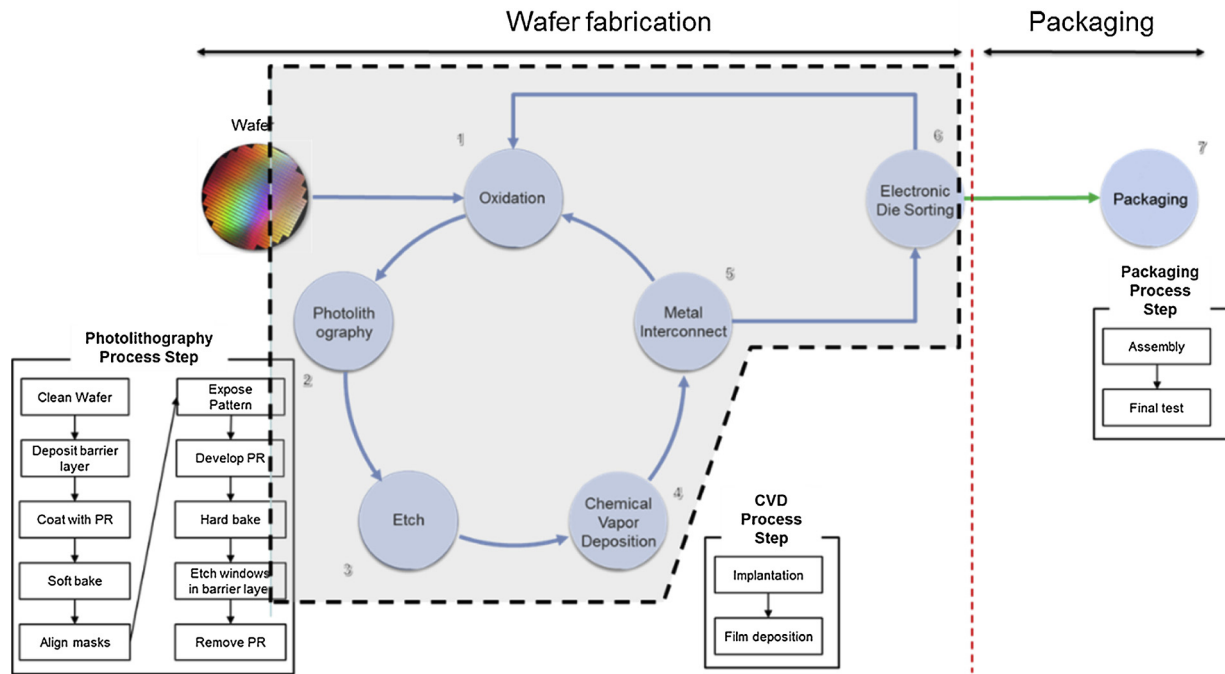


Fig. 1. Concept of unit processes and process steps in semiconductor wafer fabrication.

having measurement data for all five process steps is 0.032% ($= 0.2^5$). The random measurement policy is an effective policy in terms of statistical process control because a large number of wafers can be measured under this policy. However, this policy makes it difficult to analyze the data, especially for multivariate analysis.

The third property is the small number of low yield wafers. To investigate the effects of the process steps on wafer yield, not only high yield wafers, but also low yield wafers are required. However, as the wafer fabrication lines become mature and stable, most wafers have high yields.

These three troublesome properties are often encountered in semiconductor wafer fabrication lines. In response to the above suggested problems, we propose a systematic and data-driven approach for selecting the critical process steps. In this approach, the critical process steps are selected using a feature selection method that is well known in data mining fields. The purpose of the feature selection method is to identify key factors that significantly influence the response variable. Usually, the impact of the factors on the response variable is investigated by analyzing data, and most feature selection methods assume sufficient and structured data. However, this assumption cannot be satisfied in the suggested problem due to the three properties. To satisfy the assumption, we employ a missing value imputation method and a re-sampling method. The missing value imputation method overcomes the first and second properties, and the re-sampling method overcomes the third property.

The remainder of the paper is organized as follows. Existing methods for selecting critical process steps are reviewed and analyzed by data mining techniques such as feature selection methods, missing value imputation methods, and re-sampling methods. The proposed method is presented and illustrated with a case study. Finally, a summary and discussion are presented.

2. Literature review

2.1. Critical process step selection methods

The proposed method aims to select the critical of numerous process steps by considering the impact on wafer yield. There are similar

studies to select critical process features or variables for various manufacturing process. Kim et al. [5] developed a two-step procedure for key process variable identification and process monitoring in forging processes. In their work, globally important process variables are identified using principal component analysis. Then, localized identification of key process variables for specific quality characteristics is performed using the regression coefficient matrix and a direct clustering algorithm. Shao et al. [6] proposed a systematic procedure to select a parsimonious set of features that provides sufficient information for process monitoring. In this work, k -fold cross-validation is applied to select significant features and set monitoring limits simultaneously. Guo and Banerjee [7] applied a mapper algorithm for predictive analysis of a chemical manufacturing process for yield prediction and a semiconductor etch process for fault detection. The algorithm selects key process variables that impact system outcomes by analyzing network shape.

In addition, there exist other methods for selecting critical process steps in semiconductor wafer fabrication lines. Kang et al. [8], Susto et al. [9], and Nam and Kim [10] proposed methods that predict yield using a virtual metrology method. In virtual metrology, measurement data of each process step are predicted based on data gathered from sensors attached to machines. Then, key process steps are selected by analyzing the predicted data. The prediction by the virtual metrology corresponds to the missing value imputations. In Section 5, the proposed method is compared with the virtual metrology method in detail.

2.2. Feature selection methods

If we regard the process steps as features (independent variables) and the yield as a dependent variable, the suggested problem can be considered a feature selection problem, which is common in data mining. In this section, the feature selection methods are reviewed. Feature selection methods are used to reduce the dimension of the data space in data mining and aim to identify critical of many features. Feature selection is roughly categorized into filter and wrapper approaches [11].

The filter approach is a methodology in which features are ranked according to specific statistical criteria. Features having high values for

the statistical criteria are selected as critical. The filter approach is useful for handling high-dimensional data because the computational burden for selection is relatively small. The disadvantages are that the interactions between variables are not considered, and the number of final selected features is ambiguous [12]. The chi-square test, mutual information, and principal component analysis [13] are representative methods using a filter approach.

The wrapper approach is methodology for selecting features that maximize the performance of specific classification/regression analysis models. This approach is good in terms of the performance of the analysis models; however, it requires a large computational burden, so it is difficult to apply to high-dimensional data [12]. Sequential forward/backward selection [14], particle swarm optimization [15], and genetic algorithm [16] methods are available. There is also a number of hybrid feature selection methods that combine the filter and wrapper approaches [17–21].

Recently, Kerdprasop and Kerdprasop [22] proposed a filter approach method called ‘MeanDiff’ to improve fault detection accuracy. Given samples of independent variables (i.e., features) and a dependent variable, MeanDiff selects critical features as follows. 1) Cluster the samples into two (fail and pass clusters) based on the value of the dependent variable. 2) Compare the mean differences of every feature between the fail and pass clusters. 3) Prioritize the features in descending order according to magnitude of mean differences. Because MeanDiff is simple and effective in feature selection, we adopted MeanDiff and modified it in the proposed method. Please refer to Section 3 for the modification.

2.3. Missing value imputation methods

The problem of missing values (or missing data) is relatively common in almost all studies [23]. Missing values lead to loss of information and can cause potential bias in estimation of parameters [24]. Rubin [25] formalized three mechanisms by which missing data can occur: missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR). From a practical perspective, the mechanisms are assumptions that dictate the performance of different missing data techniques [26].

Data are MCAR when the probability of missing data on $\mathbf{X}$ (i.e., the input variables) is unrelated to other measured variables and to the values of $\mathbf{X}$ themselves. That is, missing data do not depend on observed or unobserved data. Data are MAR if they are related to other measured variables in the analysis model but not to unobserved data. It should be noted that MAR is not actually random and describes systematic missing data where the propensity for missing data is correlated with other related variables in the analysis [26]. Finally, data are MNAR when they depend on the missing values themselves.

It is difficult to identify the underlying missing mechanism in our problem. Obviously, it is not MNAR because wafers are randomly selected for measurement in each process step. We believe that the missing mechanism is close to MAR because the process steps would have correlations among them, which means that missing values would have correlations with observed values.

Traditional imputation methods such as deletion and mean imputation only produce accurate estimates when the data are MCAR. By contrast, modern techniques, such as maximum likelihood and multiple imputation, provide unbiased estimates when the data are MCAR or MAR. Popular approaches assuming MCAR or MAR are hot-deck imputation, regression imputation, expectation maximization (EM) [27], and Markov chain Monte Carlo (MCMC) [28].

The rate of missing values is directly related to the quality of statistical inferences. Yet, there is no established cutoff regarding an acceptable percentage of missing values [29]. Schafer [30] asserted that a missing rate of 5% or less is inconsequential. Bennett [31] maintained that statistical analysis is likely to be biased when more than 10% of the data are missing [29]. As we mentioned in Section 1, there is also a high

rate of missing values of operation data in the semiconductor wafer fabrication process. Therefore, we will analyze various rates of missing values in a case study.

As previously mentioned, EM is a popular method for missing value imputation. It is assumed that data follow a multivariate normal distribution with parameters $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. The EM algorithm is used for estimating $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ considering the missing values. EM is a general technique for finding the maximum likelihood estimates for parametric models when the data are not fully observed [28].

In the EM algorithm, two steps (i.e., estimation and maximization) iterate until the estimates of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ converge. Let $\mathbf{X}$ denote the $n \times p$ matrix of complete data, where n is the number of observations, and p is the number of variables. The i th observation $\mathbf{x}_i$ is a p -dimensional vector denoted by $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip})$.

$\mathbf{X}$ includes both the observed and missing data (i.e., $\mathbf{X} = (\mathbf{X}_{\text{obs}}, \mathbf{X}_{\text{miss}})$), where $\mathbf{X}_{\text{obs}}$ and $\mathbf{X}_{\text{miss}}$ denote the observed and missing data parts of $\mathbf{X}$, respectively. The purpose of the expectation step is to fill in $\mathbf{X}_{\text{miss}}$ so that the maximization step can estimate the probability model parameters $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. The expectation step builds a set of regression models that predict $\mathbf{X}_{\text{miss}}$ from $\mathbf{X}_{\text{obs}}$ and fill $\mathbf{X}_{\text{miss}}$ with the predicted values from the regression models. Then, $\mathbf{X}$ contains no missing values. Based on the new $\mathbf{X}$ (with no missing values), sufficient statistics such as $\sum \mathbf{x}_i$ are computed. When computing the sufficient statistics, the predicted values form the regression model used [26]. In the maximization step, $\hat{\boldsymbol{\mu}}^{(t)}$ and $\hat{\boldsymbol{\Sigma}}^{(t)}$ are calculated based on the sufficient statistics obtained from the expectation step. In the next expectation step, the prediction models are updated based on $\hat{\boldsymbol{\mu}}^{(t)}$ and $\hat{\boldsymbol{\Sigma}}^{(t)}$. t is an iteration counter starting with 0. Then, $\mathbf{X}_{\text{miss}}$ is updated by replacing the previous data with the new predicted data from the new prediction model. Again, new sufficient statistics are computed to calculate $\hat{\boldsymbol{\mu}}^{(t+1)}$ and $\hat{\boldsymbol{\Sigma}}^{(t+1)}$. This iterative procedure continues until $\hat{\boldsymbol{\mu}}^{(t)}$ and $\hat{\boldsymbol{\Sigma}}^{(t)}$ become the same as $\hat{\boldsymbol{\mu}}^{(t+1)}$ and $\hat{\boldsymbol{\Sigma}}^{(t+1)}$, respectively.

2.4. Re-sampling methods

Imbalanced data refer to a case where the size of one group (class) is very large or small compared to that of another group. Often, real-world datasets are predominately composed of “normal” samples (majority class) with only a small percentage of “abnormal” or “interesting” samples (minority class) [32]. This is a common problem in classification of data mining. When data are imbalanced, most classification algorithms result in poor performance [33]. The re-sampling technique can be used to resolve this problem. Re-sampling techniques are categorized into under- and over-sampling techniques.

The under-sampling technique removes samples of the majority class so that both classes have the same number of samples. This approach is simple, but there is a risk of losing valuable data. In contrast, the over-sampling technique reconstructs the sample of the minority class. Random over-sampling, the synthetic minority over-sampling technique (SMOTE) [32], modified SMOTE [34], and adaptive synthetic (ADASYN) [35] are representative over-sampling techniques. Over-sampling techniques are known to be more effective than under-sampling techniques [36] because they do not delete any data. However, the learning time is long, and an over-fitting problem may occur.

In this paper, we use SMOTE, which is one of the representative over-sampling techniques. SMOTE increases the number of minority samples by generating synthetic samples through combination of existing samples. Thus, it mitigates the problem of over-fitting by generating new samples instead of replicating samples. Also, there is no loss of useful information.

The procedure of SMOTE consists of three iterative tasks. First, one minority class sample is randomly selected and denoted as $\mathbf{x}_{\text{start}}$. Second, other k minority class samples that are close to $\mathbf{x}_{\text{start}}$ are selected by applying the k -NN (nearest neighbors). The number of selected minority class samples, k , is typically set as $k = 5$. Third, k

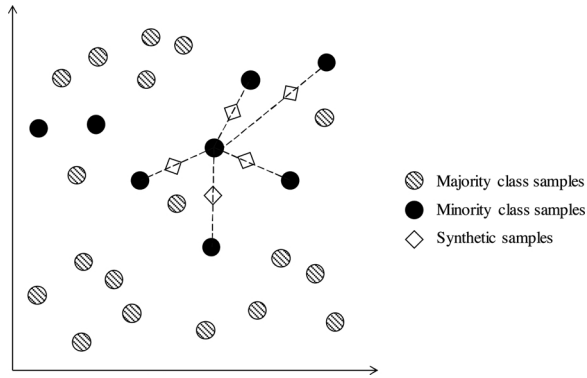


Fig. 2. Illustration of the synthetic sample generation process using the SMOTE technique.

synthetic samples are generated through randomized interpolation between $\mathbf{x}_{\text{start}}$ and the k minority class samples. Fig. 2 illustrates the generation of synthetic samples by the SMOTE technique. These three tasks are repeated until the number of minority class samples is the same as that of the majority class samples.

However, SMOTE, Modified-SMOTE and ADASYN approaches may generate incorrect or unnecessary samples in some scenarios. Recently, Zhang et al. [37] proposed a weighted minority oversampling (WMO) approach to overcome the limitations of these approaches. WMO is a new synthetic oversampling approach for imbalanced data problems in real rotating machinery applications. It consists of three steps: 1) find informative samples; 2) partition them into subspaces and determine the synthetic sample number adaptively; and 3) adopt an improved generation strategy [37]. One limitation of WMO is that it only deals with binary-class imbalanced data problems. Also, WMO requires determination of some user-specified parameters. In addition, the WMO approach has not been validated for semiconductor manufacturing processes with many features. Therefore, we employ the well-known SMOTE technique to balance the data distribution.

3. Proposed procedure

In this section, we propose a method for selecting critical process steps. A large number of process steps exist, and only the critical ones are selected through the proposed method. As previously mentioned, we consider the three properties in the proposed method. First, the missing value rate is high. Second, wafers are randomly selected for measurement in each process step. Thus, only a small amount of measurement data exist, and every process step has measurement values for different wafers. Although the measurement value has a high missing rate, the yield is inspected for every wafer; thus, the yield has no missing values. However, the number of low yield wafers is quite small compared to that of high yield wafers. Fig. 3 shows the procedure of the proposed method. Steps 2 and 3 are important in the proposed method in that they attempt to overcome the three aforementioned difficulties. Detailed explanations of the five steps are given below.

3.1. Step 1. Data cleaning

In this step, outliers and redundant features are deleted. To detect outliers, a simple 3σ rule or box plot can be used. We recommend the use of the 3σ rule if data follow a normal distribution. The 3σ rule is a simple and generally used method for outlier detection. After data cleaning, suppose that we have N wafers denoted by $\{(y_i, \mathbf{x}_i), i = 1, 2, \dots, N\}$, where y_i and $\mathbf{x}_i$ are the yield value and vector of measurement values at the process steps (i.e., features) of the i th wafer, respectively. $\mathbf{x}_i$ is a p -dimensional vector denoted by $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip})$, where p denotes the number of process steps, and x_{ij} is the measurement value at the j th process step of the i th wafer. Usually, several response variables

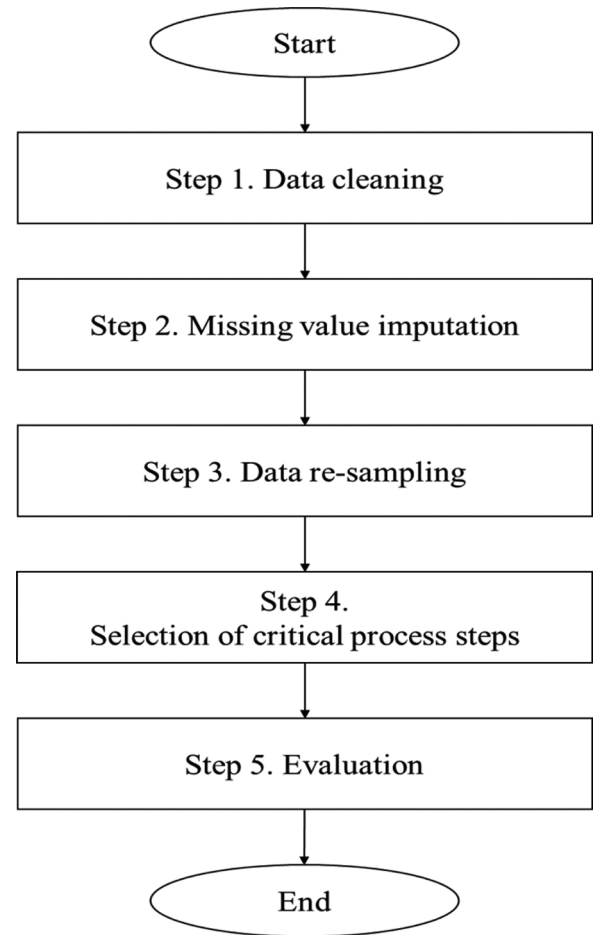


Fig. 3. Overall procedure of the proposed method.

are measured at each process step, and x_{ij} is a main response variable among them. We assume that y is a binary variable, so y_i has a value of 0 or 1; $y_i = 0$ indicates that the i th wafer is a low yield wafer.

3.2. Step 2. Missing value imputation

In this step, missing values are imputed. We recommend the use of the EM algorithm introduced in Section 2. EM is recommended because the missing data mechanism of the suggested problem satisfies the assumption on the missing data mechanism of the EM algorithm. In the suggested problem, the missing value occurs randomly under random sampling inspection policy, corresponding to MCAR (missing completely at random) or MAR (missing at random). Also, the EM algorithm is a general method for missing value imputation and takes into account the relationships between other variables; thus, it could be adequately applied to the proposed method.

3.3. Step 3. Data re-sampling

Usually, the number of low yield wafers indicated by $y_i = 0$ is small. To overcome the imbalance issue, we recommend use of the SMOTE technique, which is one of the representative over-sampling techniques. SMOTE requires small computational effort, which is important to the suggested problem because the number of features (i.e., process steps) is typically large (i.e., more than 500 features) in the suggested problem. This technique generates artificial low yield wafers at a number the same as that of wafers whose $y_i = 1$.

3.4. Step 4. Selection of critical process steps

As previously mentioned in Section 2.1, we use the modified MeanDiff technique to select the critical features. For the same reason as for SMOTE, we employ MeanDiff because it requires small computational effort. Wafers are classified into two clusters, fail and pass clusters, and differences of process operational data between the two clusters are compared. Wafers of fail and pass clusters have y values of 0 and 1, respectively. $\bar{x}_{0j}$ and $\bar{x}_{1j}$ are the average values of the j th feature of fail and pass clusters, respectively.

$$\bar{x}_{0j} = \frac{\sum_{i=1}^N x_{ij}(1 - y_i)}{\sum_{i=1}^N (1 - y_i)}, j = 1, 2, \dots, p,$$

$$\bar{x}_{1j} = \frac{\sum_{i=1}^N x_{ij}y_i}{\sum_{i=1}^N y_i}, j = 1, 2, \dots, p.$$

Similarly, s_{0j}^2 and s_{1j}^2 are the sample variance of the j th feature of fail and pass clusters, respectively.

$$s_{0j}^2 = \frac{\sum_{i=1}^N (x_{ij}(1 - y_i) - \bar{x}_{0j}(1 - y_i))^2}{(\sum_{i=1}^N (1 - y_i)) - 1}, j = 1, 2, \dots, p,$$

$$s_{1j}^2 = \frac{\sum_{i=1}^N (x_{ij}y_i - \bar{x}_{1j}y_i)^2}{(\sum_{i=1}^N y_i) - 1}, j = 1, 2, \dots, p.$$

Once the four statistics for $j = 1, 2, \dots, p$ are computed, T_j is computed as follows from the well-known two sample t -test. Then, all of the features are ranked in descending order according to T_j value.

$$T_j = \frac{|\bar{x}_{0j} - \bar{x}_{1j}|}{\sqrt{\frac{s_{0j}^2}{\sum_{i=1}^N (1 - y_i)} + \frac{s_{1j}^2}{\sum_{i=1}^N y_i}}}$$

When comparing the modified MeanDiff with the original MeanDiff, the modified MeanDiff considers variability of the feature data, whereas the original MeanDiff only considers the mean difference of the two clusters (i.e., $|\bar{x}_{0j} - \bar{x}_{1j}|$). The modified MeanDiff can be considered a scale-free approach because it uses a standardized value T_j . Therefore, when scales of features are different, we recommend the use of the modified MeanDiff for a fair comparison.

3.5. Step 5. Evaluation

Finally, the selected features are evaluated. For this purpose, we build a prediction model in which the wafer is classified by the selected features. Then, evaluation is conducted based on the performance of classification.

Four terms are used for evaluating the performance of the prediction models, and they are shown in Fig. 4. The TP rate is the rate of fail

samples that are correctly classified. A higher TP rate means better performance. The FP rate is the rate of pass samples that are wrongly classified. A lower FP rate means better performance. Precision represents the detectability of fail samples. A higher value of Precision means better performance. The F-measure calculates the harmonic mean of the Precision and the TP rate. A higher value means better classification performance for the fail class.

4. Case study

In this section, the proposed method is illustrated via a case study by analyzing the SECOM dataset. This dataset is obtained from a semiconductor manufacturing process and is available on the UCI machine learning repository [38]. In this case study, we assume that the feature corresponds to the semiconductor wafer fabrication process step. The SECOM dataset contains 1567 samples of 590 features. Each sample corresponds to a wafer, and every wafer has its yield as binary data: pass or fail. y values of the pass and fail wafers are $y_i = 1$ and $y_i = 0$, respectively. There are 1463 pass wafers and 104 fail wafers, which means the dataset is imbalanced.

In this case study, we intentionally modify the SECOM dataset to reflect the three properties frequently encountered in the semiconductor wafer fabrication process. The dataset is imbalanced; thus, the third property is already reflected in the dataset. The missing rate of the original dataset is approximately 5%. To reflect the first and second properties, we randomly delete 80% of the data for each feature.

This section consists of four subsections. First, we select the critical process steps (i.e., critical features) according to the procedure explained in Section 3 but do not perform Steps 2 and 3. Secondly, we conduct the procedure again by including Steps 2 and 3. We do this to investigate the effect of Steps 2 and 3, which are the main focus of the proposed method for resolving the difficulties of the suggested problem. In the third and fourth subsections, sensitivity analyses with respect to a missing value rate are conducted.

4.1. Case study without Steps 2 and 3

The case study is conducted by analyzing the modified SECOM dataset. As previously mentioned, 80% of the data are randomly deleted for each feature, and the analysis is conducted with the remaining 20% of the data. In Step 1, we perform data cleaning for the modified SECOM dataset. There are no outliers based on the 3σ rule. Second, 120 features having the same value across the entire sample are removed. Table 1 shows the dataset before and after the data cleaning step.

We do not perform Steps 2 and 3. In Step 4, we use the modified MeanDiff technique to select the critical features. As a result, 47 features that are in the top 10% of the entire set of features are selected as the critical process steps. In Step 5, we build a prediction model in

		Predicted class		
		fail	pass	Total
Actual class	fail	TP	FN	P
	pass	FP	TN	N
Total		P'	N'	P + N

$$accuracy = \frac{TP + TN}{P + N}$$

$$TP\ rate = \frac{TP}{P}$$

$$FP\ rate = \frac{FP}{N}$$

$$Precision = \frac{TP}{P'}$$

$$F-measure = \frac{2TP}{2TP + FP + FN}$$

Fig. 4. Confusion matrix and equations for performance measurement.

Table 1
Numbers of features and samples before and after Step 1.

	Before Step 1	After Step 1
# of features	590	470
# of pass samples	1463	1463
# of fail samples	104	104

which the wafers are classified by the selected critical process steps. A 25% holdout cross-validation is used to build the prediction model using the training set. That is, the prediction models are fitted to 75% of samples, and the other 25% of samples are used for validating the fitted models. We used four algorithms to build a model: decision tree, logistic regression, k -nearest neighbors (k -NN), and support vector machine (SVM). The result of the confusion matrix is shown in Table 2.

Because the case study is conducted without Steps 2 and 3, all wafers have at least one missing value for the selected critical process step. In such a case, SVM and logistic regression cannot generate prediction model. Both SVM and logistic regression classify the wafers with prediction models built by estimating coefficients for the selected critical process steps and using them as input variables. In SVM, the coefficients are estimated by solving the optimization problem to maximize the margin between hyperplanes. In logistic regression, the coefficients are estimated by maximizing a likelihood function. When at least one missing value exists, the coefficients cannot be estimated, and prediction models cannot be generated.

In contrast, k -NN classifies wafers without prediction models. According to k -NN, a wafer is classified based on the k nearest wafers in the learning samples. This mechanism is available when distances between wafers can be calculated, and this calculation is possible even with missing data. For example, suppose a simple case of two wafers having [1,5] and [2, 2,] for three features. The first and second wafers have missing values for the second and third features, respectively, and both have observed data for the first feature. Then, their distance is calculated as 1 based on the first feature. In this way, k -NN classification is available with missing values; however, the calculated distance can be misleading because it comes only from the observed values. Thus, the classification performance is poor, as shown in Table 2.

On the other hand, none of the fail wafers can be predicted. Because the dataset is imbalanced, all of the wafers are predicted as the pass class. This shows that, without Steps 2 and 3, the prediction models have poor performance.

4.2. Case study with Steps 2 and 3

In this section, the modified SECOM dataset is analyzed by including Steps 2 and 3. The result of Step 1 is the same as in the Section 4.1. In Step 2, we perform the EM algorithm to impute the missing values. As a result, the missing rate is 0%. In Step 3, data re-sampling is performed using the SMOTE technique. Table 3 summarizes the data changes so far.

In Step 4, we select 47 critical features that are in the top 10% of the entire set of features. Lastly, we generate the prediction models in the same manner as in the previous section. The results of the confusion matrix are shown in Table 4 and Fig. 5. The results show much better classification performance than those without Steps 2 and 3.

Table 2
Confusion matrix for the four models without Steps 2 and 3.

Decision tree		Predicted class		Logistic regression, SVM		Predicted class		k -NN		Predicted class	
		Fail	Pass			Fail	Pass			Fail	Pass
Actual class	Fail	0	26	Actual class	Fail	0	0	Actual class	Fail	0	26
	Pass	1	364		Pass	0	0		Pass	0	365

Table 3
Data information before and after each step.

	Before Step 1	After Step 1	After Step 2	After Step 3
# of features	590	470	470	470
# of pass samples	1463	1463	1463	1463
# of fail samples	104	104	104	1463
Missing rate	80%	80%	0%	0%

Because no missing values exist, all of the performance indices can be calculated, whereas they could not be in Section 4.1. In the results, accuracy is better in the order of SVM, k -NN, decision tree, and logistic regression. SVM has the best accuracy and F-measure values compared to the other models. Overall, SVM shows good performance compared to the other models.

Fig. 5 compares the performance indices from Table 2 (without Steps 2 and 3) and Table 3 (with Steps 2 and 3). In Table 2, the number of pass samples is much larger than the number of fail samples (i.e., 365 pass samples and 26 fail samples). Due to this imbalanced data, all wafers were classified as pass wafers. Although the accuracy is high (i.e., $0.936 = 365 / 391$), TP rate is 0 and is not available to compute precision. In contrast, when Steps 2 and 3 are conducted, all of the performance indices are computed, and their scores are reasonably good.

4.3. Effects of missing value rates

We investigate the effects of missing value rates in this section. For this purpose, we consider three data deletion rates (i.e., 80%, 50%, and 20%). We randomly delete 80% of the data for each feature from the original SECOM dataset three times to obtain three datasets. Similarly, we randomly delete 50% and 20% of the data three times, respectively, to obtain six datasets (three for 50% data deletion and three for 20% data deletion). For each dataset, 47 critical process steps are selected by the proposed method. As a result, we obtain nine sets of 47 critical process steps, which can be considered a one-factor design with three replicates.

Table 5 reports average values of performance indices of SVM and k -NN for the three deletion rates. Accuracy is an overall score of classification. When the four indices record good scores, accuracy also records a good score. k -NN records a slightly better TP rate than SVM; however, its FP rate, precision, and F-measure are much worse. As a result, k -NN shows less accuracy than SVM.

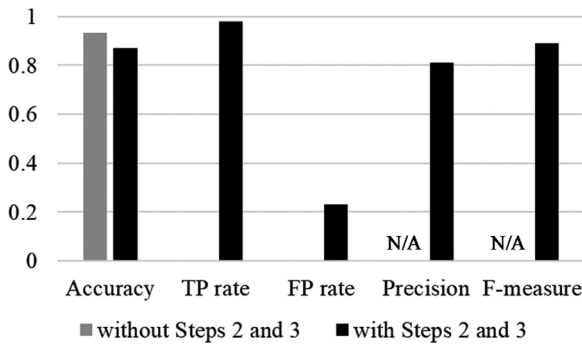
k -NN records high TP rate and low precision because it tends to predict wafers as fail wafers. This tendency increases Type I error while decreasing Type II error (Type I and II errors are the risk of misjudgment for pass and fail wafers, respectively). Also, this tendency produces a low FP rate because many pass wafers are classified as fail wafers (i.e., Type II error).

Compared to k -NN, SVM records low TP rate and high precision because the effect of missing value imputation is different for the two methods. In other words, k -NN is more sensitive to missing value imputation than SVM, as shown in Table 5. The effects of missing value imputation are different because the mechanisms of SVM and k -NN are different. SVM builds the prediction model for classification, and the prediction model is determined by a small set of samples called

Table 4

Confusion matrix for the four models with Steps 2 and 3.

Decision tree				Logistic regression				k-NN				SVM			
		Predicted class				Predicted class				Predicted class		Predicted class		Predicted class	
		Fail	Pass			Fail	Pass			Fail	Pass			Fail	Pass
Actual class	Fail	201	165	Actual class	Fail	249	117	Actual class	Fail	363	3	Actual class	Fail	346	20
	Pass	20	345		Pass	90	275		Pass	122	243		Pass	37	328

**Fig. 5.** Comparison of the performance indices of *k*-NN from Tables 2 and 4.

“Support Vectors.” In contrast, *k*-NN classifies a wafer based on nearest samples and searches nearest samples for every wafer to be classified. Compared to SVM, more samples have a chance to participate in the prediction process in *k*-NN, producing high sensitivity to the missing value rate.

Tables 6 and 7 report the ANOVA results of performance indices for SVM and *k*-NN, respectively. SVM shows stable performances for the three deletion rates. Although 80% of the data are deleted, the proposed method works well compared to the 20% data deletion case. In *k*-NN, the performance becomes better as the deletion rate decreases, and the improvement is significant at $\alpha = 5\%$, except that of TP rate, as shown in Table 7. As previously mentioned, due to its mechanism, *k*-NN is more dependent on missing value imputation than SVM.

Overall, the result of the analysis implies that SVM is best in this case study. It works well when the missing value rate (i.e., deletion rate) is small and shows stable performance with the high missing value rate.

4.4. Robustness with respect to random deletion trials

As mentioned in Section 4, we randomly delete data to reflect the first and second properties. Random deletion has an effect on selection of the critical steps. To investigate this effect, we compare the nine sets of selected critical process steps obtained in Section 4.3. Table 8 shows the nine sets of selected critical process steps.

For the 80% deletion case, 15, 12, and 11 features are common between 1st and 2nd, 1st and 3rd, and 2nd and 3rd random deletion trials, respectively. The average common portion rate is 27% ($= (38/47)/3$). In this way, we assess the number of common features between three random deletion trials for the other deletion rates. Fig. 6 shows the average portion of common features for the three deletion rates. As the deletion rate decreases, more features are commonly selected. Table 9

reports ANOVA results for number of common features according to deletion rate. The effect of deletion rate was statistically significant, implying that the proposed method is stable when the missing value rate is small.

5. Discussion

In Section 2.1, existing methods for selecting the critical process steps are introduced. Both the proposed and existing methods attempt to identify critical variables in a manufacturing process. However, none of the existing methods considers all three properties explained in Section 1 that are often encountered in semiconductor wafer fabrication lines. The proposed method employs missing value imputation and re-sampling methods to handle the three troublesome properties.

Table 10 compares the proposed method with the existing methods ([5–10]) in terms of the target process, missing value imputation, and re-sampling. In terms of the target process, only the virtual metrology methods [8,6–10] target the semiconductor fabrication process as does the proposed method. In terms of missing value imputation, only the virtual metrology methods conduct the missing value imputation process as does the proposed method. In terms of re-sampling, none of the existing methods conduct the re-sampling process; thus, they do not consider the imbalanced data issue. Overall, the virtual metrology methods are similar to the proposed method.

It is interesting to compare the mechanism of missing value imputation of the virtual metrology methods with that of the proposed method. The proposed method imputes the missing values by analyzing the relationships among the measurement variables of process steps. In contrast, the virtual metrology methods predict the missing values of the measurement variables of process steps based on other source of data, called machine data, which are gathered from the sensors attached to machines. Assume that $\mathbf{x}$ and $\mathbf{z}$ are the vectors of the measurement variables and the machine variables, respectively. Due to the first property explained in Section 1, $\mathbf{x}$ has a high rate of missing values. In contrast, $\mathbf{z}$ is expected to have no missing values because the machine data are automatically gathered from sensors. The proposed method imputes the missing values by building prediction models among the measurement variables in an iterative manner and do not require $\mathbf{z}$. In contrast, the virtual metrology methods use $\mathbf{z}$ to build a prediction model, $\hat{\mathbf{x}} = f(\mathbf{z})$, and impute the missing values of $\mathbf{x}$ through $f(\mathbf{z})$. There exist several practical difficulties in virtual metrology. The performance of virtual metrology methods depends on the capability of the prediction model; however, it is often difficult to maintain the high prediction capability of the prediction model. In practice, the machine condition changes frequently and $\mathbf{z}$ changes dramatically whenever the machine condition changes. The prediction model needs to be updated whenever the machine condition changes to maintain the high

Table 5Average values of performance indices of SVM and *k*-NN for the three deletion rates.

Deletion rate	SVM					k-NN				
	Accuracy	TP rate	FP rate	Precision	F-measure	Accuracy	TP rate	FP rate	Precision	F-measure
80%	0.89	0.95	0.16	0.86	0.90	0.82	0.99	0.35	0.74	0.85
50%	0.90	0.95	0.14	0.87	0.91	0.85	0.99	0.28	0.78	0.87
20%	0.88	0.92	0.16	0.86	0.88	0.87	0.98	0.23	0.81	0.89

Table 6
ANOVA of performance indices of SVM.

Source	Df	Accuracy				TP rate				FP rate				Precision			
		SS	MS	F	P-value	SS	MS	F	P-value	SS	MS	F	P-value	SS	MS	F	P-value
Deletion rate	2	0.00	0.00	1.30	0.34	0.00	0.00	1.48	0.30	0.00	0.00	0.32	0.74	0.00	0.00	0.55	0.60
Error	6	0.00	0.00			0.00	0.00			0.01	0.00			0.00	0.00		
Total	8	0.00				0.00				0.01				0.00			

prediction capability. However, this is often difficult in practice because the machine changes frequently and non-periodically, and there are large numbers of machines and variables to be monitored. Also, z is often noisy compared to x , because of instable operation of the sensors and unstable transmission of the machine data. For these reasons, the proposed method is easier to use and more practical because it does not build the prediction models from the machine data.

In a real-world semiconductor manufacturing line, data are gathered from various sources such as sensors from machines, measurement machines, inspection machines, and humans. Data from sensors are often noisy due to unstable operation or unstable transmission of data. Data types are different depending on the data source. In the suggested problem, we assumed that only quantitative data exist with no noisy data. For better real-world applications, various data types should be considered. For example, data from human inspection are often text data in words. In such a case, text mining methods can be applied for the proposed method. One of the applicable text mining methods is word2vec, which is an open source tool based on deep learning that can learn the vector representations of the words in the high-dimensional vector space and calculate a similarity index such as cosine distance between words [39]. In the proposed method, word2vec can be used to calculate the similarity index between words associated with fail and pass wafers in Step 4. That is, the distance measure in Step 4 of the proposed method (i.e., T_j) can be replaced with the similarity index. Then, the critical features can be selected based on the calculated similarity index.

Another data type that is often encountered in real-world situations is qualitative data. Data from measurement machines are often quantitative, while data from inspection machines are often qualitative. The data from the inspection machines are qualitative in that the inspection machines investigate whether each semiconductor chip in a wafer is defective or not. Although the chips judged defective can be revived through subsequent process steps, these chips are likely to fail in the final test. When applying these qualitative data to the proposed method, the proportions of defective chips for the pass and fail wafers need to be calculated. Then, similar to the text data case, the distance measure in Step 4 of the proposed method (i.e., T_j) can be modified by a new test statistic for two proportions of fail and pass wafers.

On the other hand, we assumed that only one feature exists for each process step. In a real-world semiconductor manufacturing line, many features can exist for each process step. In such a case, a hypothetical feature such as principal component for each process step can be used to represent the process step.

In Step 4 of the proposed method, the MeanDiff technique that uses t -statistics is used for selecting the critical process steps and does not take into account the interactions between the process steps. Process

faults are often propagated in terms of metrologies of consecutive steps. For example, the results after the Litho process will be propagated to the Etch and then to CMP. By simply comparing the t -statistics, the origin of the issues can be missed.

From a practical point of view, it is difficult to investigate the interaction effects for the entire process steps because the number of process steps is often large (i.e., more than 500). This requires tremendous data and computational efforts. Moreover, when main effects of factors are not significant, their interaction effects are also often not significant. In this regard, we suggest investigating the interaction effects only for a small number of process steps. The process steps with a T_j value in Step 4 could be included for investigation. The interaction effects among them can be estimated by adding the interaction terms to prediction models and fitting the prediction models to data. This shows the improved performance of the prediction model.

In Step 5, yield prediction models are built to evaluate the selected features. We believe that over-fitting can be systematically prevented by the proposed method. First, we employ SMOTE (synthetic minority over-sampling technique) as the over-sampling method to mitigate the over-fitting problem by generating new samples instead of replicating samples. Second, the number of input variables in the prediction models is quite small compared to the number of samples. Usually, as the prediction model includes more input variables, it is likely to be over-fitted. As an extreme example, if the number of input variables is the same as the number of training samples, then the model can perfectly predict the response of the training samples. However, for other samples such as test samples, large prediction error would occur. In the proposed method, a large number of samples is used for building the yield prediction model. In addition, the number of input variables is quite small because only the selected critical features are used as the input variables in the yield prediction model.

In practice, the critical features (i.e., critical process steps) are often selected based on the knowledge of domain experts in practice. Usually, the domain knowledge is valuable and helpful to select the critical features. However, it is often difficult to provide evidence of domain knowledge quantitatively. We expect that most of the features selected by domain experts will be also selected by the proposed method, and vice versa. In this regard, the proposed method can be used to support the domain expert's opinion by providing quantitative evidence. Also, when the domain experts do not have enough knowledge of the process (this often happens when new processes are developed), the set of critical features selected by the proposed method can serve as a reference for the domain experts to select the critical features.

Table 7
ANOVA of performance indices of k -NN.

Source	df	Accuracy				TP rate				FP rate				Precision			
		SS	MS	F	P-value	SS	MS	F	P-value	SS	MS	F	P-value	SS	MS	F	P-value
Deletion rate	2	0.00	0.00	11.10	0.01	0.00	0.00	0.39	0.69	0.02	0.01	11.07	0.01	0.01	0.00	11.77	0.01
Error	6	0.00	0.00			0.00	0.00			0.01	0.00			0.00	0.00		
Total	8	0.01				0.00				0.03				0.01			

Table 8
Selected features with respect to deletion rate and random deletion trial.

Deletion rate	Random deletion trial	Selected features
80%	1 st	X12, X18, X24, X29, X31, X41, X54, X55, X60, X78, X80, X90, X122, X123, X124, X125, X126, X128, X130, X154, X188, X222, X255, X280, X289, X291, X296, X299, X320, X342, X349, X352, X412, X421, X426, X432, X456, X469, X489, X500, X511, X556, X558, X560, X562, X584
	2 nd	X22, X29, X39, X42, X60, X64, X79, X81, X89, X96, X104, X122, X123, X124, X125, X128, X130, X131, X134, X153, X159, X161, X164, X171, X181, X198, X245, X288, X296, X299, X317, X342, X427, X432, X442, X453, X487, X500, X501, X511, X543, X544, X546, X548, X558, X576, X578
	3 rd	X10, X12, X22, X26, X27, X38, X39, X44, X61, X67, X71, X79, X80, X122, X123, X128, X130, X133, X145, X147, X163, X181, X182, X196, X205, X213, X248, X298, X299, X300, X325, X351, X353, X356, X366, X412, X421, X476, X477, X489, X520, X546, X561, X562, X575
50%	1 st	X15, X22, X29, X32, X60, X64, X65, X66, X77, X87, X104, X112, X122, X124, X125, X130, X131, X144, X160, X164, X176, X182, X184, X240, X248, X282, X295, X299, X300, X317, X431, X432, X435, X436, X437, X442, X447, X453, X454, X456, X489, X511, X520, X546, X574, X576, X578
	2 nd	X12, X20, X22, X29, X32, X39, X60, X64, X80, X122, X123, X124, X125, X126, X127, X130, X131, X134, X138, X148, X161, X164, X181, X111, X214, X248, X271, X273, X281, X283, X289, X296, X299, X349, X388, X399, X431, X432, X435, X436, X437, X447, X469, X488, X511, X512, X520, X546
	3 rd	X22, X29, X32, X34, X60, X64, X79, X104, X116, X122, X123, X124, X125, X126, X128, X130, X131, X161, X181, X184, X188, X250, X281, X296, X299, X300, X317, X320, X349, X388, X431, X432, X435, X436, X437, X442, X453, X456, X461, X511, X520, X543, X548, X574
20%	1 st	X22, X29, X39, X59, X60, X64, X65, X66, X80, X91, X104, X112, X122, X123, X124, X125, X126, X128, X130, X131, X134, X161, X164, X181, X211, X240, X245, X247, X281, X282, X296, X299, X317, X349, X383, X384, X385, X431, X432, X435, X436, X437, X453, X469, X511, X512, X519
	2 nd	X22, X29, X34, X38, X58, X60, X64, X65, X80, X86, X104, X112, X122, X123, X124, X125, X126, X128, X130, X131, X158, X161, X164, X181, X211, X248, X281, X282, X288, X293, X296, X299, X317, X349, X426, X431, X432, X435, X436, X437, X453, X469, X500, X511, X520, X566, X568, X570
	3 rd	X10, X22, X29, X32, X34, X60, X64, X65, X66, X71, X77, X79, X80, X86, X101, X104, X122, X123, X124, X125, X126, X128, X129, X130, X131, X134, X161, X181, X211, X248, X296, X299, X317, X349, X386, X431, X432, X435, X436, X437, X453, X469, X511, X512, X520, X543

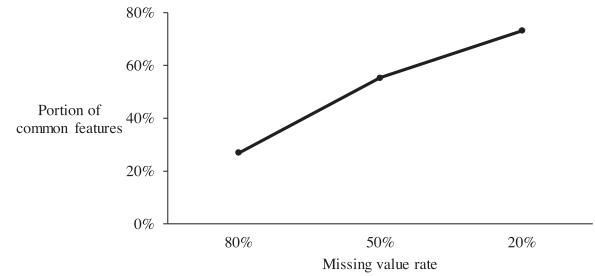


Fig. 6. Average portion of common features for three missing value rates.

Table 9
ANOVA for the number of common features according to deletion rate.

Source	df	SS	MS	F	P-value
Deletion rate	2	716.67	358.33	64.5	0.000087
Error	6	33.33	5.56		
Total	8	750			

6. Concluding remarks

Semiconductor wafer fabrication consists of hundreds of complex processes. Among them, some process steps have a strong relationship with wafer yield, and these steps are called “critical process steps.” The critical steps should be carefully selected and managed because they greatly affect wafer yield. However, the selection criteria are often ambiguous and depend on the experience and intuition of the process engineers.

Recently, it has become possible to collect and analyze a huge amount of data in real time, and a methodology for analyzing such a large amount of data is being actively studied. In this paper, we proposed a systematic and data driven approach for selecting the critical process steps. With a recent development of data analytics technologies, semiconductor companies are attempting to identify the critical process steps by analyzing the huge amount of process operational data. The proposed method is meaningful for this recent trend in semiconductor industry. The proposed method considers three properties of the data from the wafer fabrication process by adopting data mining methods, such as the data imputation method, re-sampling method, and feature selection method. As a case study, we used the SECOM dataset to investigate the effect of the proposed method under various missing value rates. As shown in Table 6, the proposed method showed good classification performance metrics for various deletion rates.

We suggest four future research directions. First, the robustness of the proposed method needs to be improved. As in Section 4.4, the proposed method is sensitive to random deletion trials when the missing value rate is high. Other various missing data imputation methods need to be investigated in our method for better robustness. Second, evaluation criteria for the selected critical steps need to be developed. Indicators for equipment health assessment can be applied for this purpose [40]. Third, we need to consider continuous values for yield data. In this paper, we used discrete values for yield (0 or 1) because the SECOM dataset has discrete values for yield. However, yield has continuous values ranging from 0 to 1 in practice. By considering this point, more realistic analysis can be conducted. Fourth, the proposed method can be extended further by considering the real-world issues in a semiconductor manufacturing line such as noisy data, various data types, and interaction effects among the process steps.

Conflict of interest

Nothing declared.

Table 10
Comparison with the existing methods for identifying critical variables in a manufacturing process.

	Proposed method		Kim et al. [5]		Shao et al. [6]		Guo and Banerjee [7]		Virtual metrology (Kang et al. [8]; Susto et al. [9]; Nam and Kim [10])	
	Target process	Semiconductor fabrication process	Forging process	General manufacturing process	General manufacturing process	General manufacturing process	General manufacturing process	Semiconductor fabrication process	Semiconductor fabrication process	Semiconductor fabrication process
Missing value imputation	○	○	×	×	×	×	×	○	×	×
Re-sampling	○	○	×	×	×	×	×	○	×	×

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